Inorganic Cumulative Exam

December 4, 2015

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded.

1. $AlF_3$ and $BF_3$ are isoelectronic and would be expected to have similar properties. Draw the Lewis dot formulation for each and use the formulations to predict a reaction that these two compounds would undergo.
2. Use an MO diagram to show the location of the unpaired electron in NO.
3. Ferrocene is sometimes used as an internal standard in cyclic voltammetry. The electrochemical reaction that is used as a reference has a potential of 0.64 V vs SHE and is said to be reversible. What is the electrochemical reaction and by what criteria is it considered reversible?
4. The crystal structures of many salts can be rationalized in terms of a cubic close packing or hexagonal close packing model. Describe these two models.
5. Organometallic hydrides such as $HCo(CO)_4$ behave as protic acids yet they are called d-metal hydrides. Are both of these descriptions consistent with the 18 electron rule? Explain.
6. Many metals have moderately low melting points but very high boiling points. For example, aluminum melts at 660 °C and boils at 2467 °C whereas bromine, a nonmetal, melts at -8 °C and boils at 58 °C. Rationalize this trend using any bonding theory.
7. One of the commonly used entry points in organometallic synthesis is photochemical loss of a CO. Give an example and provide one advantage to using this approach.
8. $[Cr(OH_2)_6]^{3+}$ ions are pale blue in solution but the chromate ion $CrO_4^{2-}$ is an intense yellow. Characterize the transitions and explain the relative intensities.
9. Boron nitride, $(BN)_3$ is a compound that illustrates the power of isoelectronic comparisons remarkably well. Provide an isoelectronic comparison to boron nitride and indicate how it illustrates the predictive power of the comparison. If you are not familiar with the specific example make some predictions.
10. Hemoglobin is the oxygen carrier found in blood. The active center is Fe(II) which reversibly binds $O_2$. Many transition metal complexes including those of Fe(II) bind oxygen readily but they rapidly convert to irreversibly bound oxygen dimers. In light of this problem how would you try to synthesize a synthetic oxygen carrier?